

Module 2: Recommender Systems

Follow-up + Evaluation

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Assignments

- Overall, good responses
 - *Please put your group # in the submission and both names
- Quite a few poor descriptions of your solutions
 - What did you actually implement?
- Mostly good recommendations from your systems!
 - Many of you didn't try to evaluate w.r.t. e.g., dimension of SVD
- Many noted that evaluation is hard—but not exactly why

Retrieval

Take 3–5 minutes and write down the most important points from the introduction lecture on recommendation systems.

2. How would you **evaluate** a system?

Optimized objective

- We tried minimizing the **empirical risk of predicting ratings**

$$\hat{L}_{\text{tr}}(\theta) := \sum_{j=1}^m \frac{1}{n_j} \sum_{i:r(i,j)=1} (\theta_i x_j - y_{ij})^2$$

User preferences for i
Rating

$r(i,j) = 1$
 if user i rated item j

Number of observed ratings of j
Item features for j

- A natural approach to evaluation is to look at our achieved objective. What was our error in prediction?

Generalization error in "standard" ML

- We want to be accurate also for unrated items, so we can recommend them if we think users might like them!
- I.e., our goal could be to reduce **expected risk** in a distribution p

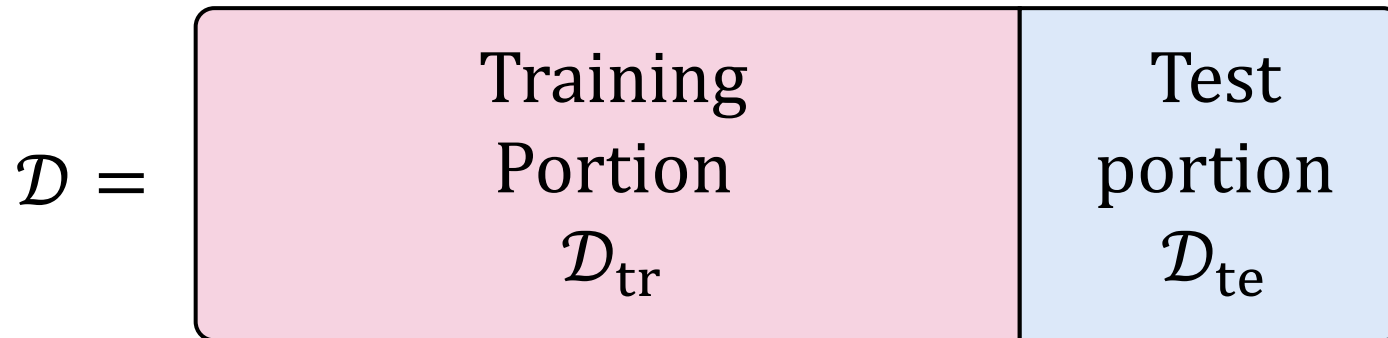
$$L_p(\theta) := \mathbb{E}_{X,Y \sim p} [(\theta^\top X - Y)^2]$$

Expected error in predictions (irrespective of whether a user has rated or not)

- $L_p(\theta) - \hat{L}_{\text{tr}}(\theta)$ is called the **generalization error**

Train/test splits

- Typically, in machine learning, we split our data and evaluate a fitted model $\hat{\theta}$ on the test portion



- Why do we do this?

Independent samples

Training
Portion
 $\mathcal{D}_{\text{tr}}$

Test
portion
 $\mathcal{D}_{\text{te}}$

$\hat{\theta}$ and $\mathcal{D}_{\text{tr}}$ are dependent!

$$\hat{\theta} = \min_{\theta} \frac{1}{n_{\text{tr}}} \sum_{i \in \mathcal{D}_{\text{tr}}} \ell_{\theta}(x_i, y_i)$$

$\hat{\theta}$ and $\mathcal{D}_{\text{te}}$ are independent!

$$\hat{L}_{\text{te}}(\hat{\theta}) \approx L_{\textcolor{blue}{p}}(\theta)$$

Is this always true?

Generalization in recommendation

- But then what is the distribution of X, Y in our case?

- For a single user i ,

Distribution of item features and opinions

$$L_i(\theta) := \mathbb{E}_{X, Y_i}[(\theta^\top X - Y_i)^2] = \int_{x, y} \ell_\theta(x, y) \overset{|}{p(\mathbf{x}, \mathbf{y})} dx dy$$

- In data, we only see rated movies! Samples from $p(X, Y_i \mid R_i = 1)$!
- This is true also for the test set. Hence, $\hat{L}_{\text{te}}(\theta) \not\approx L_{\mathbf{p}}(\theta)$

Distributional shift

- Rated movies $p(X, Y \mid R = 1)$ are a **biased subset** of movies
- What's worse, the distribution $p(X, Y, R)$ is **not fixed**! We have:
 - New users (generalization error)
 - New viewing patterns (selection bias)
 - ...
- We observe data under $\hat{p}(X, R, Y)$ and predict under $q(X, Y)$
- This is prediction under **distributional shift**!

Theme: “Empirical risk minimization”

- Finding a model $\hat{\theta}$ that minimizes error on observed data

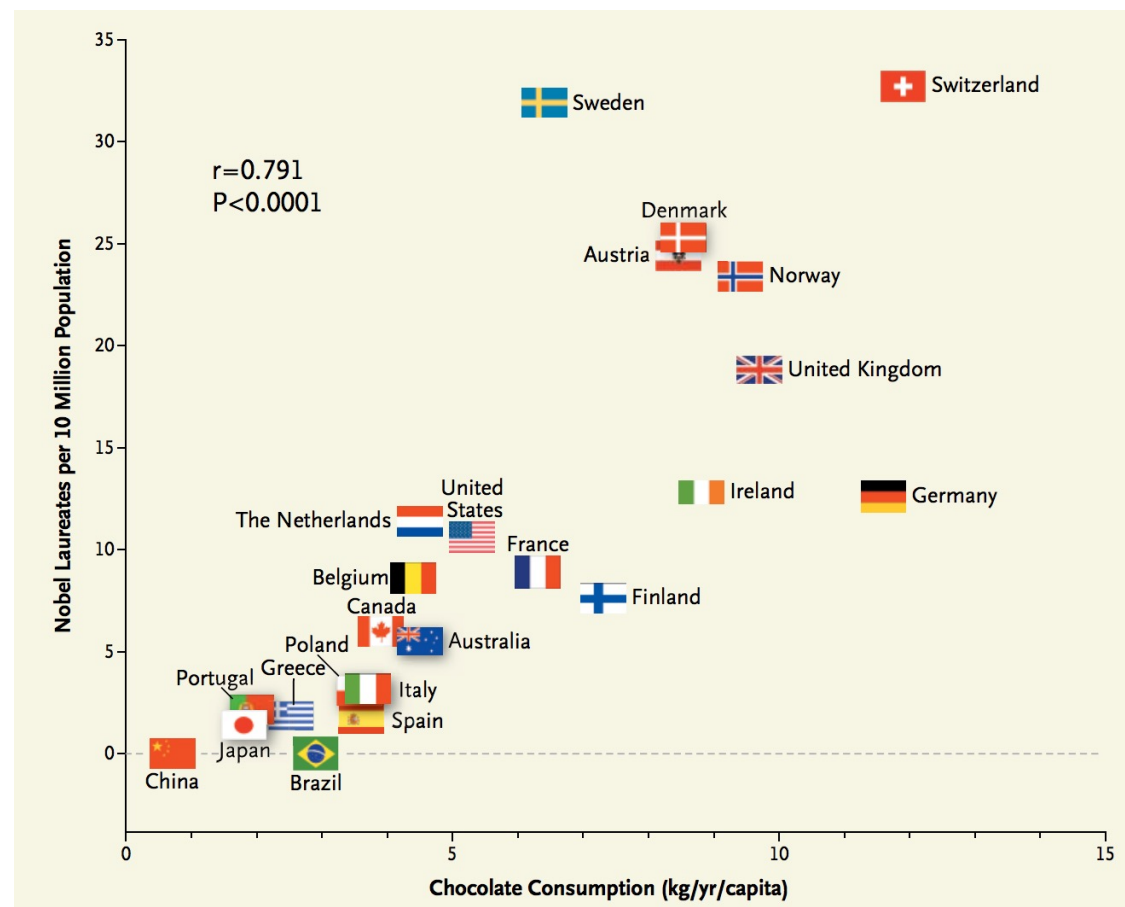
$$\hat{\theta} = \arg \min_{\theta} \sum_{j=1}^m \sum_{i:r(i,j)=1} (\theta_i x_j - y_{ij})^2$$

- By far the most common AI/ML strategy today
- But assumes that data is sampled from the “test” distribution!

Consistent learning

- When we assume that $\hat{\theta}$ is a good model on p , we usually implicitly assume that $\hat{L}_{\text{tr}}(\theta) \rightarrow L_p(\theta)$ when $n_{\text{tr}} \rightarrow \infty$
- For many models, this implies $\hat{\theta} \rightarrow \arg \min_{\theta} L_p(\theta)$.
This is called “consistent” learning
- When $p_{\text{tr}} \neq p$, we don’t typically have this guarantee

Correlation & causation



Correlation & causation

More on this in later weeks!

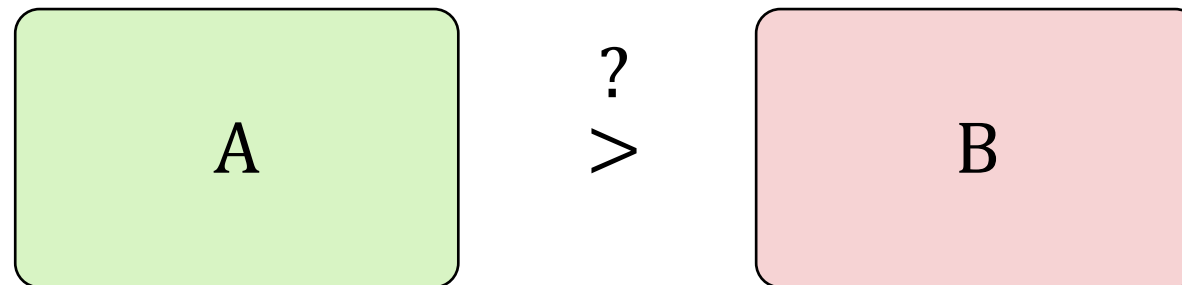
Recall: Objectives & proxies

- Our ability to predict may be good, but for a **bad system**
- For example, $\mathbb{E}_{X,Y}[(\theta^\top X - Y)^2 \mid R = 1]$ may be low even if $Y \approx 0$
- Prediction error is one thing, but we really care about the **ratings**
- We want to **make Y go up** by changing our recommendations!
 - (And hence, what is rated, R)

A/B Tests

Online A/B experiments

- An A/B test compares two systems (A & B) at the same task



- With online services, user bases may be split at any time and **selected into experiments**
- **Example:** Half of users get recommendations from A, half from B

Online A/B experiments

- Experiment setting
 - Select a user segment
 - Allocate traffic to different models
- Result reporting
 - Report experimental results
 - Impact on a many metrics

in

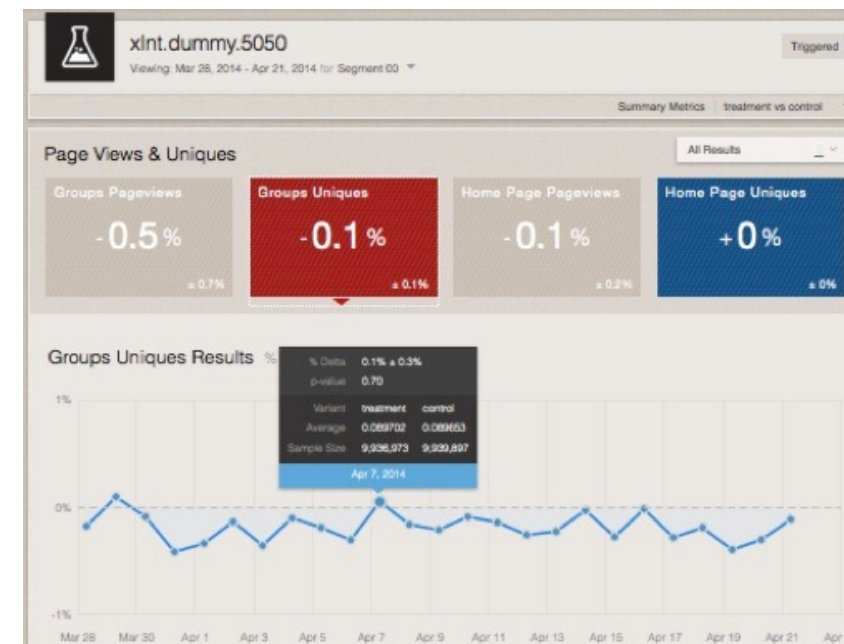
country-code

us,cn

Allocation

50% treatment

50% control



¹Slides from Deepak Agarwal, <https://www.youtube.com/watch?v=VJOtr47V0eo>

Online & offline A/B tests

Online tests

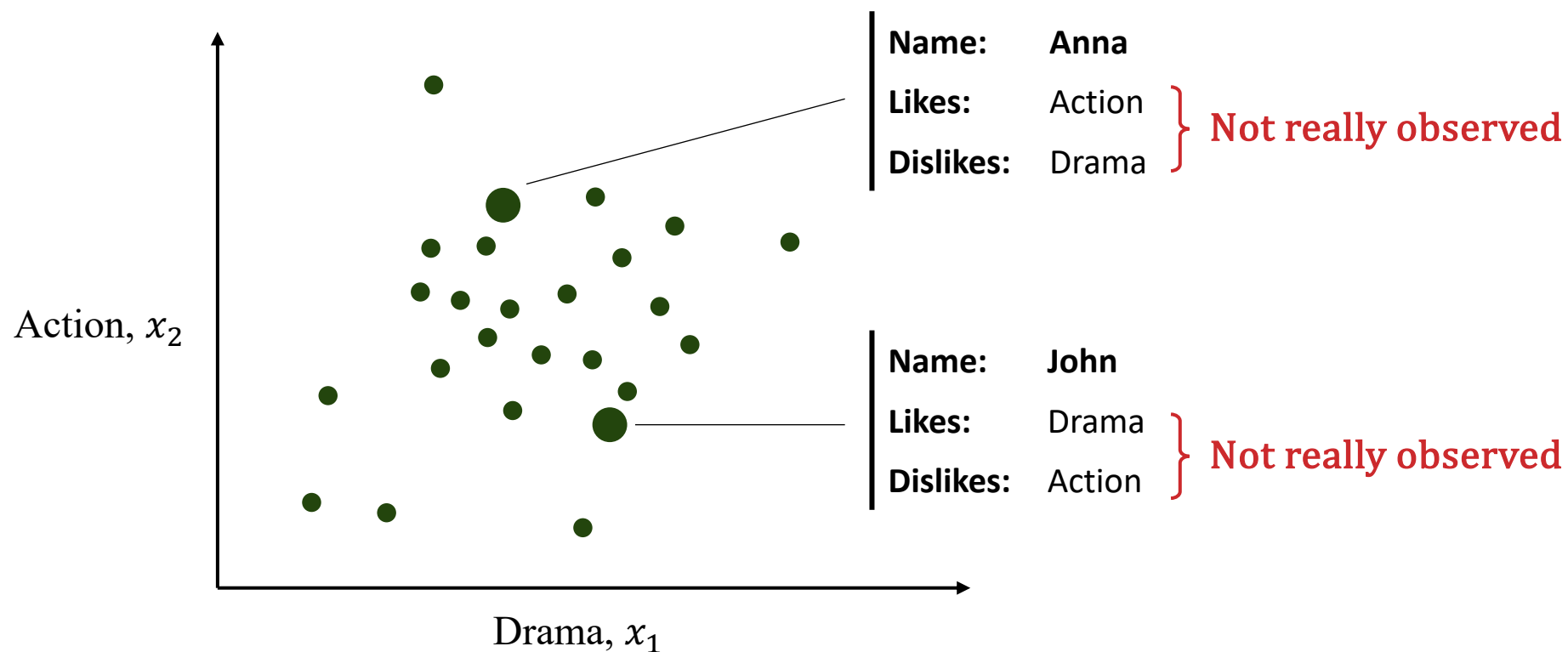
- The new system is used in production, side-by-side with the old one
- Costly, but less sensitive to selection bias

Offline tests

- Knowledge from the old system is used to predict behavior under the new one
- Has to compensate for **selection bias** (old policy affected which items are rated)

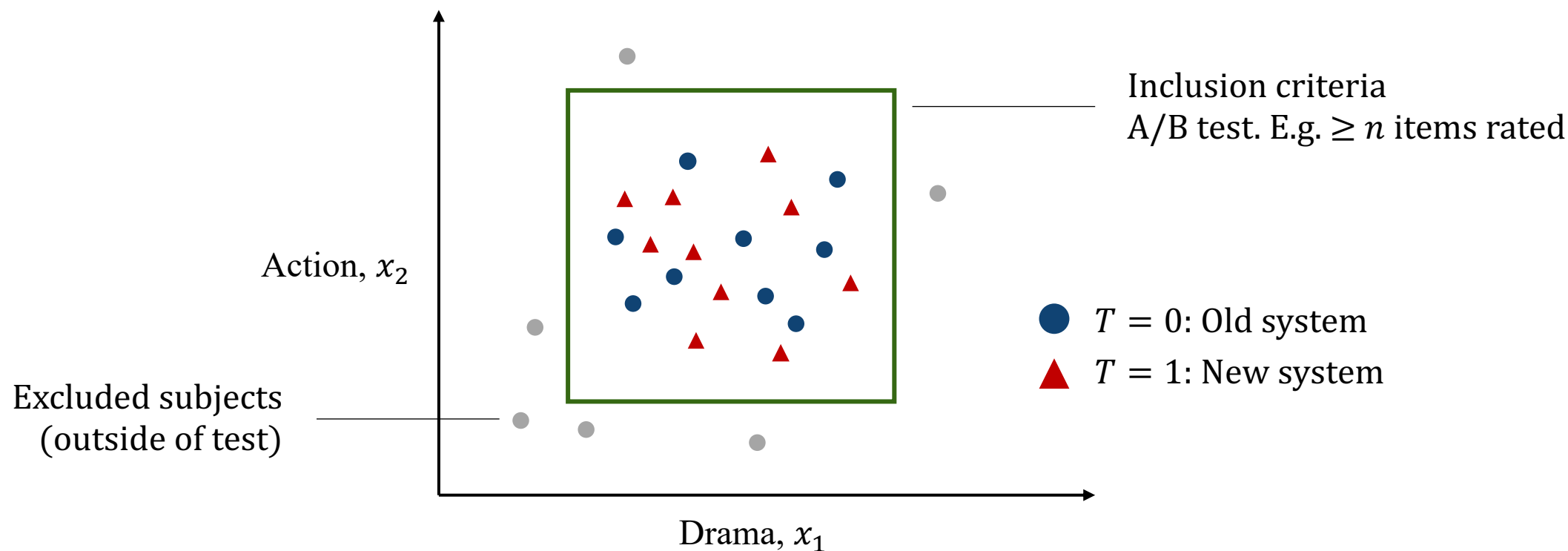
Online A/B Test

1. Candidate population/users are identified



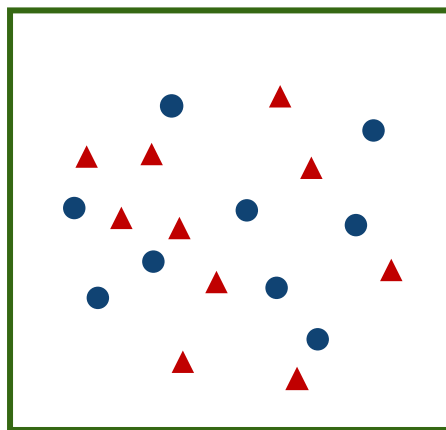
Online A/B Test

2. Subjects are filtered and allocated “treatment” randomly



Online A/B Test

- We can estimate the effect of the new system



● $T = 0$: Old system
 ▲ $T = 1$: New system

Average causal
effect

$ACE \approx$

Average rating
under new system

$$\frac{1}{n_1} \sum_{i:t_i=1} y_i$$

Collected under $q(x, y)$

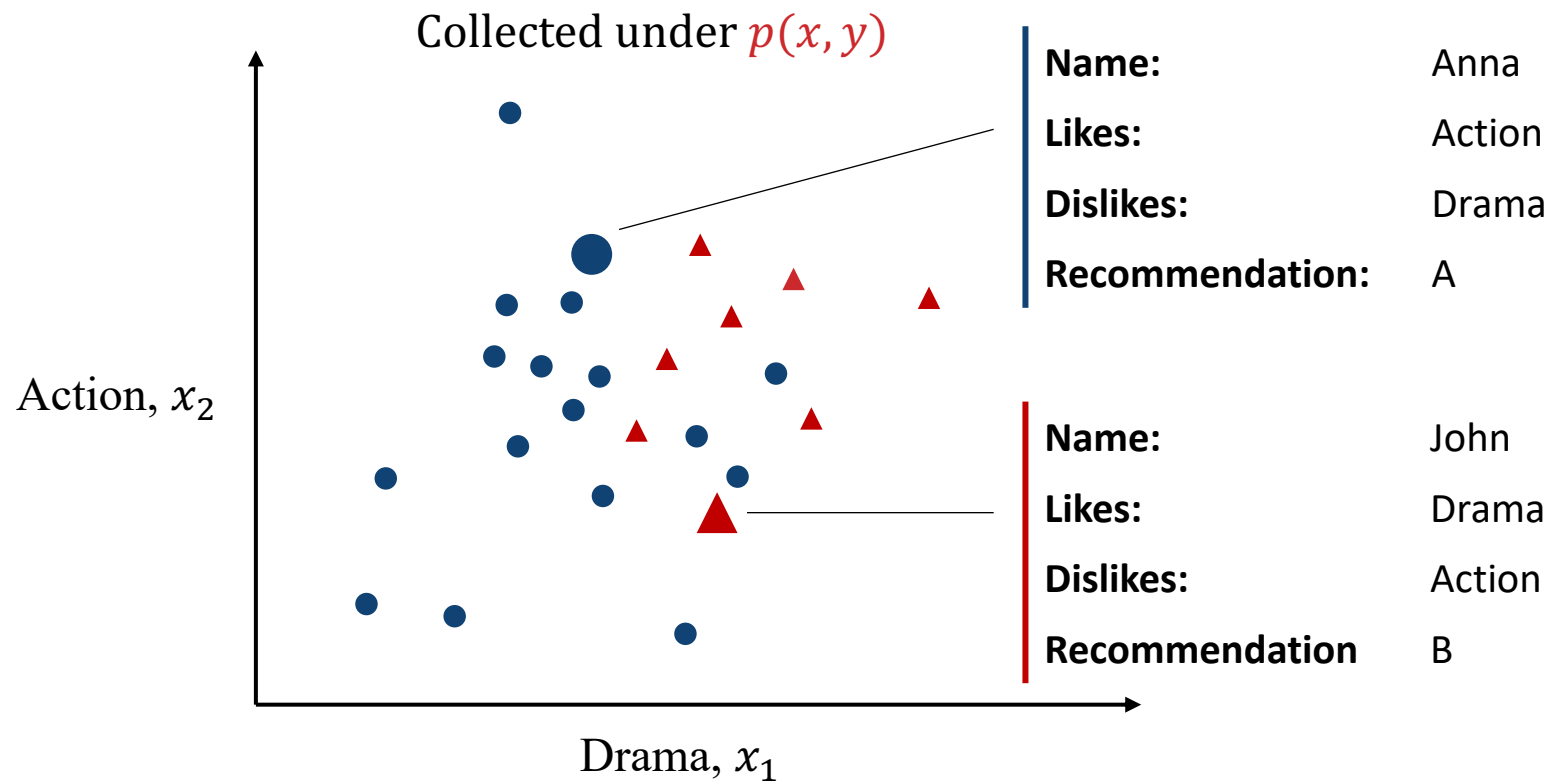
Average rating
under old system

$$- \frac{1}{n_0} \sum_{i:t_i=0} y_i$$

Collected under $p(x, y)$

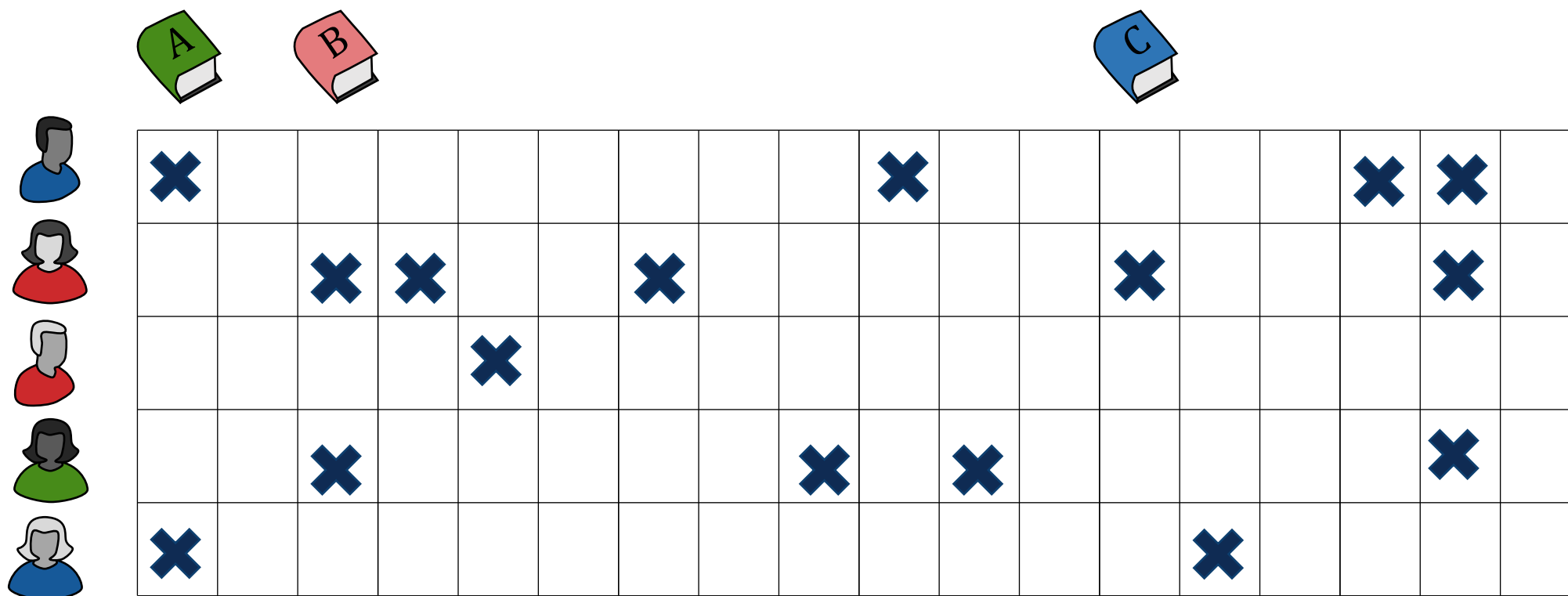
Offline A/B tests

1. Recommendations have already been assigned



Missing at random

- If users selected movies randomly, this would be OK



Missing at random

- If users selected movies randomly, this would be OK
- Then, selection and high ratings are **uncorrelated!**
- But if selection was random, our recommendation were **very bad**

	A	B														
User 1	X								X						X	X
User 2			X	X			X				X					X
User 3					X											
User 4			X						X	X						X
User 5	X												X			

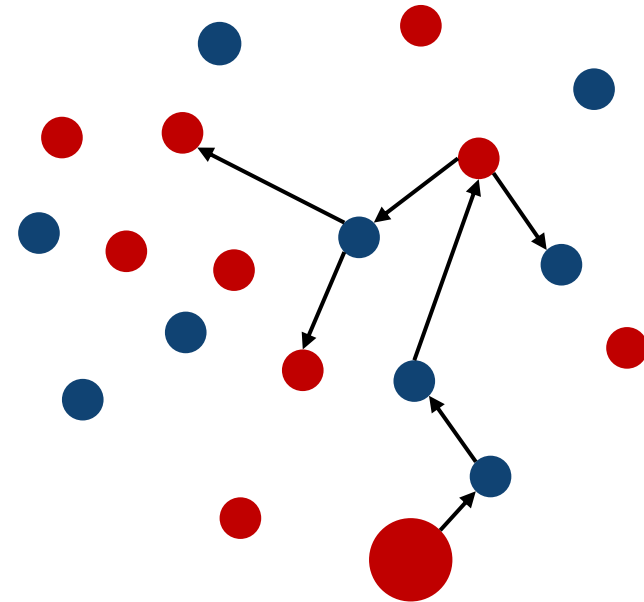
4. What can go wrong?

Discussion portion

- I will ask a **series** of questions
- You will **discuss these** in groups
- In the end, a random group will report answers to **each of these**
 - Prepare answers to each!
 - You're welcome to use points from your assignments! Share!
- I will “walk” around and take part/listen in

Network effects

- Users may **influence each other**. What one user is exposed to may influence what others are exposed to
- **Discuss:** How can this affect your system's accuracy? What assumption (implicit or explicit) does this violate? Can you adjust for it?



What happens when a new product is introduced?

- In some cases, you have no features on new products. Even worse, none of your users has interacted with them
- **Discuss:** How should **new products** be handled? Pick one application of recommender systems and discuss the question w.r.t. that application

Evaluation under non-stationarity

- Your users will change their behavior over time
- **Discuss:** What could cause such change? How can changes in **user taste, context** affect your system? How does it affect our assumptions about $p(x, y)$? How can we address this?

Offline A/B tests

Discuss: How could you use “offline” data to estimate the error/ratings under a new policy? How could you account for the **selection bias**? What assumptions do you make?

